Rishik Vamshi Rohith Vempati

ai24mtech12003@iith.ac.in | rishikvempati@gmail.com | +91 9493019049 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.76
FIITJEE JUNIOR COLLEGE	Class XII (Telangana State Board Of Intermediate Education)	2019	92.80%
DR K.K.R'S GOWTHAM CONCEPT SCHOOL	Class X (Board Of Secondary Education, Telangana)	2017	9.20%

RESEARCH EXPERIENCE

Under the guidance of Dr. Konda Reddy Mopuri

- **Investigated Test-Time Adaptation (TTA)** literature across **episodic, batch, and online** settings; focused on **episodic TTA** and developed computationally and memory-efficient approaches, contributing to our **CVPR paper**.
- **Currently exploring Topological Data Analysis (TDA)** at the intersection of **Test-Time Adaptation and Computer Vision**, investigating topology-aware approaches for robust adaptation.

PUBLICATIONS

Research work "**IMA & TMA: Efficient Test-Time Adaptation for VLMs via Linear Transformation in Embedding Space**" accepted at **CVPR** "*ViSCALE Test-Time Scaling for Computer Vision*" workshop.

SKILLS

- **Programming Languages** : C, Java, Python, JavaScript, SQL
- **Technologies & Frameworks** : Spring Boot, PyTorch, Optuna, LangChain
- **Developer Tools** : Git, DagsHub
- **Areas of Interest** : Data Structures & Algorithms (DSA), Computer Vision (CV), Generative AI, LLM Inference

PROJECTS

Industry-Sponsored Project | Advisor: Dr. Konda Reddy Mopuri

*ML Tool for **3D Point Cloud Classification** of Automotive Components*

- Engineered end-to-end deep learning pipelines to process raw 3D point cloud scans of internal automotive assemblies.
- Leveraged PointNet++ and PV-RCNN, collaborated with industry engineers to optimize preprocessing and performance.
- Addressed data and annotation challenges, delivering an industry-ready tool with **>95% classification accuracy**.

Swiggy Delivery Time Prediction

- **Engineered a feature pipeline** for geospatial anomaly correction, categorical temporal feature extraction, applying Power Transformation on skewed target and missing-data handling; tracked experiments using **MLflow** on **DagsHub**.
- Performed hyperparameter optimization and model selection using **Optuna** across **six regression algorithms**; isolated and fine-tuned **LightGBM and Random Forest** models for improved predictive performance.
- Architected a **Stacking Regressor** meta-model combining the optimized LightGBM and Random Forest base models, achieving **0.85 train R-squared score** and **0.86 test R-squared score**.

Movie Recommendation System

- Built a **Movie GraphRAG** pipeline using **Gemini, LangChain, and Neo4j** to extract movie knowledge from PDFs and construct a relationship-aware knowledge graph.
- Integrated **Neo4j graph search** with **Pinecone vector search** to enable natural-language movie queries and semantic movie recommendations.

SCHOLASTIC ACHIEVEMENTS

- **GATE CSE 2024 Qualified** -- Top **0.3**% among 1.23 Lakh+ Candidates
- **NPTEL Programming in Java** -- Awarded **Gold Badge (**Top **5%)** with a consolidated score of **91%**

RELEVANT COURSES

- **Core Coursework**: Matrix Theory & Linear Algebra, Probability & Random Variables, Stochastic Processes, Convex Optimization, Foundations of Machine Learning, Deep Learning, Data Structures & Algorithms
- **Online Certifications & Training** : **CCNA v7**: **Introduction to Networks** (*Cisco Networking Academy*), **Workshop on Basic Python Programming** (*FOSSEE project, IIT Bombay*), **Python Data Structures** (*Coursera, University of Michigan*), **Google Cloud Digital Leader Training** (*Coursersa, Google Cloud*)

POSITIONS OF RESPONSIBILITY

Technical Instructor -- AIET, IIT Hyderabad × Accenture

- Taught **Computer Vision** to **40+ industry professionals** as part of the **Artificial Intelligence in Emerging Technologies** program, bridging theoretical concepts with practical applications in image classification.

Teaching Assistant

- Taught core **Machine Learning** concepts and guided students through **coding assignments and practical implementations**; conducted problem-solving sessions for **Matrix Theory**, mentoring students on theoretical concepts and applications.

Student Organizer -- India AI Pre-Summit

- Served as a student organizer for the **Summit**, held at IIT Hyderabad's Conclave on Inclusion for Social Empowerment.

OCS Student Coordinator

- Coordinated placement activities and **pitched IIT Hyderabad talent to prospective companies**; co-designed the official placement brochure and led targeted outreach to **15+ prospective corporate partners.**